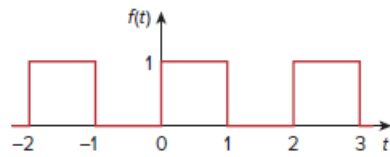


Problem

Obtain the complex Fourier series of the function in Fig. 17.1.

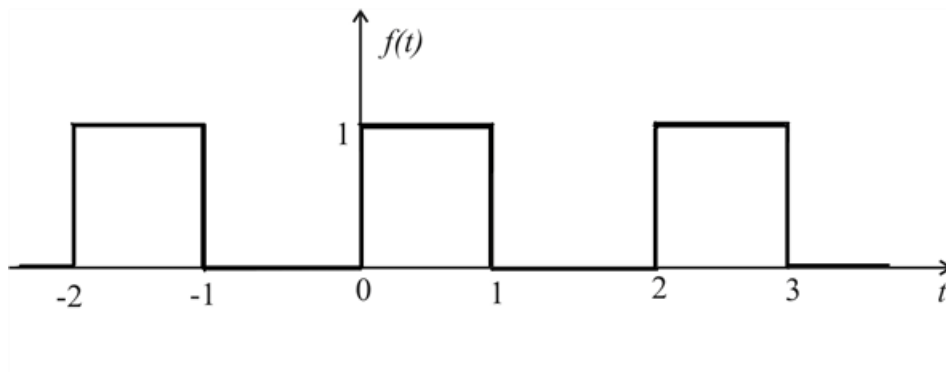
Figure 17.1:



Step-by-step solution

Step 1 of 5

Given square wave is,



Step 2 of 5

Thus, the function is

$$f(t) = \begin{cases} 1, & 0 < t < 1 \\ 0, & 1 < t < 2 \end{cases}$$

Since $T = 2$, $\omega_0 = \frac{2\pi}{T}$

$$\omega_0 = \frac{2\pi}{2}$$

$$\omega_0 = \pi$$

Step 3 of 5

$$C_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$C_0 = \frac{1}{2} \int_0^1 (1) dt$$

$$C_0 = \frac{1}{2} t \Big|_0^1$$

$$\therefore C_0 = \frac{1}{2}$$

Step 4 of 5

$$C_n = \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt$$

$$C_n = \frac{1}{2} \int_0^1 (1) e^{-jn\pi t} dt$$

$$C_n = \frac{1}{2} \left[\frac{e^{-jn\pi t}}{-jn\pi} \right]_0^1$$

$$C_n = \frac{-1}{j2n\pi} [e^{-jn\pi} - 1] \quad \text{since } e^{-jn\pi} = e^{jn\pi} = (-1)^n$$

$$C_n = \frac{-1}{j2n\pi} [(-1)^n - 1]$$

$$C_n = \frac{-1}{j2n\pi} [-2] \text{ where } n \text{ is odd}$$

$$\therefore C_n = \begin{cases} \frac{1}{jn\pi} & n \text{ is odd} \\ 0 & n \text{ is even} \end{cases}$$

Step 5 of 5

Therefore the complex fourier series of the function is,

$$f(t) = C_0 + \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$f(t) = \frac{1}{2} - \sum_{\substack{n=-\infty \\ n \neq 0 \\ n=\text{odd}}}^{\infty} \frac{j}{n\pi} e^{jn\pi t}$$